

A Continuum of Evolution-Generator Extensions of One Evolution Pre-Generator

Automated mathematics run `fa_banach_001`, lane 1

August 9, 2026

Status. Candidate full solution, likely valid, pending expert review. A single canonical evolution pre-generator associated with a constant closed operator on $L^2(0,1)$ has continuum many pairwise distinct evolution-generator extensions. Equivalently, the corresponding correctly posed problem has continuum many solution operators in the weak sense of the source paper.

1 Source question

Neidhardt, Stephan, and Zagrebnov [1] work on $L^p(\mathcal{I}, X)$, where $\mathcal{I} = [0, T]$, $T > 0$, $1 \leq p < \infty$, and X is a separable Banach space. Their Definition 3.1 calls an operator $\mathcal{K}$ an *evolution pre-generator* if its domain consists of continuous functions, is invariant under multiplication by $W^{1,\infty}(\mathcal{I})$ scalar functions, satisfies

$$\mathcal{K}M(\phi)f - M(\phi)\mathcal{K}f = M(\dot{\phi})f,$$

and has a dense cross-section in X at every $t \in (0, T]$. It is an *evolution generator* if, in addition, it generates a strongly continuous semigroup under the paper's sign convention $e^{-\tau\mathcal{K}}$.

For a constant closed operator C on X , let $\mathcal{C}$ denote its induced multiplication operator on $L^p(\mathcal{I}, X)$, and let

$$D_0f = f', \quad \text{dom}(D_0) = \{f \in W^{1,p}(\mathcal{I}, X) : f(0) = 0\}.$$

The canonical evolution pre-generator is

$$\tilde{\mathcal{K}} = D_0 + \mathcal{C}, \quad \text{dom}(\tilde{\mathcal{K}}) = \text{dom}(D_0) \cap \text{dom}(\mathcal{C}). \tag{1}$$

In Remark 3.6(i), on PDF page 14 (printed page 12), the paper asks:

“It is an open problem whether an evolution pre-generator admits several extension which are evolution generators.”

The spelling is transcribed exactly. The remark adds that such an example would give a non-autonomous Cauchy problem with more than one solution operator under Definition 3.5.

Now, we precise the notion of the *solution operator* of the problem (1.1) versus the *propagator* $\{U(t, s)\}_{(t,s) \in \Delta}$ on the Banach space X that we first described in Introduction §1.

Definition 3.5.

- (i) The evolution non-ACP (1.1) is called *correctly posed* in $\mathcal{I}_0 = \mathcal{I} \setminus \{0\}$ if $\tilde{\mathcal{K}}$ defined by (3.2) is an evolution pre-generator.
- (ii) A propagator $\{U(t, s)\}_{(t,s) \in \Delta}$ is called a *solution operator* of the correctly posed evolution problem (1.1) if the corresponding evolution generator $\mathcal{K}$ (Theorem 3.3) is an operator extension of $\tilde{\mathcal{K}}$, i.e. $\tilde{\mathcal{K}} \subseteq \mathcal{K}$.
- (iii) The evolution problem (1.1) has a unique solution operator if $\tilde{\mathcal{K}}$ admits only one extension that is an evolution generator.

Remark 3.6.

- (i) It is an open problem whether an evolution pre-generator admits several extension which are evolution generators. However, if this is case then the non-ACP (1.1) has more than one solution operator.
- (ii) Our Definition 3.5 (ii) of a correctly posed non-ACP is a *weak* property. For example, the notion of well-posedness developed in [18] implies this property.

To find extensions of the evolution pre-generator $\tilde{\mathcal{K}}$ which are evolution generators is, in general, a nontrivial problem. However, there is a special case, that easily guarantees the existence of such extension and, moreover, it is unique.

Theorem 3.7. *Assume that the non-ACP (1.1) is correctly posed in $\mathcal{I}_0$. If the evolution pre-generator $\tilde{\mathcal{K}}$ is closable in $L^p(\mathcal{I}, X)$ and its closure $\mathcal{K}$ is a generator, then the evolution problem (1.1) has a unique solution operator.*

Proof. Assume that $\mathcal{K}$ belongs to the class $\mathcal{G}(M, \beta)$. Then by Lemma 2.16 of [15] the estimate

$$\|f(t)\|_X \leq \frac{M}{(\xi - \beta)^{(p-1)/p}} \|(\mathcal{K} + \xi)f\|_{L^p}, \quad f \in \text{dom}(\mathcal{K}),$$

holds a.e. in $\mathcal{I}$ for all $\xi > \beta$. In particular, one gets for any $f \in \text{dom}(\tilde{\mathcal{K}})$:

$$\|f\|_C \leq \frac{M}{(\xi - \beta)^{(p-1)/p}} \|(\tilde{\mathcal{K}} + \xi)f\|_{L^p}.$$

Hence, we conclude for the closure $\mathcal{K}$ of $\tilde{\mathcal{K}}$ one has $\text{dom}(\mathcal{K}) \subset C(\mathcal{I}, X)$.

Now, we show that $\mathcal{K}$ is an evolution generator. Let $f \in \text{dom}(\mathcal{K})$. Then, by the closeness of $\mathcal{K}$, there

Figure 1: Definition 3.5, Remark 3.6, and the immediate uniqueness context in Neidhardt–Stephan–Zagrebnov [1], PDF page 14.

2 Proof intuition

The question is an extension problem, so begin with a familiar closed operator having many semigroup-generator extensions. The closed minimal Laplacian on $(0, 1)$ is contained in every Robin Laplacian. Each nonnegative Robin realization is self-adjoint and hence generates a contraction heat semigroup. Passing from a spatial heat semigroup to the associated evolution semigroup on $L^p([0, T], L^2(0, 1))$ turns every Robin extension into an evolution-generator extension of the same canonical pre-generator.

The dense-cross-section requirement does not destroy this construction. At any positive time, a smooth scalar cutoff which vanishes at time zero can be multiplied by an arbitrary vector from the dense domain of the minimal Laplacian. Finally, different Robin parameters impose different boundary conditions, so their spatial generators, propagators, and evolution generators are pairwise distinct. Since the Robin parameter ranges over an interval, the one pre-generator has continuum many extensions.

3 Full answer

Let $H = L^2(0, 1)$. Define the closed minimal Laplacian

$$Su = -u'', \quad \text{dom}(S) = H_0^2(0, 1) = \{u \in H^2(0, 1) : u(0) = u'(0) = u(1) = u'(1) = 0\}. \quad (2)$$

Equivalently, S is the closure of $-d^2/dx^2$ initially defined on $C_c^\infty(0, 1)$.

Theorem 1. *Fix $T > 0$ and $1 \leq p < \infty$. Let $\mathcal{S}$ be the multiplication operator on $L^p([0, T], H)$ induced by the closed minimal Laplacian S , and set*

$$\tilde{\mathcal{K}} = D_0 + \mathcal{S}, \quad \text{dom}(\tilde{\mathcal{K}}) = \text{dom}(D_0) \cap \text{dom}(\mathcal{S}).$$

Then $\tilde{\mathcal{K}}$ is an evolution pre-generator. Moreover, it has continuum many pairwise distinct closed extensions which are evolution generators. Consequently, the constant problem

$$\dot{u}(t) = -Su(t)$$

has continuum many solution operators in the sense of Definition 3.5(ii) of [1].

Proof. The operator S is closed and densely defined on the separable Hilbert space H . Its induced multiplication operator is

$$\text{dom}(\mathcal{S}) = \{f \in L^p(\mathcal{I}, H) : f(t) \in \text{dom}(S) \text{ a.e. and } Sf(\cdot) \in L^p(\mathcal{I}, H)\}.$$

We first verify the pre-generator property. Since $\text{dom}(\tilde{\mathcal{K}}) \subset \text{dom}(D_0)$, every domain element has a continuous representative. For $\phi \in W^{1,\infty}(\mathcal{I})$, scalar multiplication preserves both $\text{dom}(D_0)$ and $\text{dom}(\mathcal{S})$, and the Leibniz rule gives

$$\tilde{\mathcal{K}}M(\phi)f - M(\phi)\tilde{\mathcal{K}}f = M(\dot{\phi})f.$$

It remains to check the dense cross-section. Fix $t_0 \in (0, T]$, $x \in H$, and $\varepsilon > 0$. Because $\text{dom}(S)$ is dense in H , choose $u \in \text{dom}(S)$ with $\|u - x\|_H < \varepsilon$. Choose $\psi \in C^\infty([0, T])$ with $\psi(0) = 0$ and $\psi(t_0) = 1$. Then $f(t) = \psi(t)u$ lies in $\text{dom}(\tilde{\mathcal{K}})$ and $\|f(t_0) - x\|_H < \varepsilon$. Thus every positive-time cross-section is dense. This also recovers directly the hypotheses of Lemma 3.4 of [1].

For each $a \in [0, \infty)$, consider the densely defined nonnegative closed form

$$q_a(u, v) = \int_0^1 u'(x) \overline{v'(x)} dx + a(u(0) \overline{v(0)} + u(1) \overline{v(1)}), \quad \text{dom}(q_a) = H^1(0, 1). \quad (3)$$

The first representation theorem for closed forms [2, Chapter VI] gives a nonnegative self-adjoint operator A_a . Integration by parts identifies it as the Robin Laplacian

$$A_a u = -u'', \quad \text{dom}(A_a) = \{u \in H^2(0, 1) : u'(0) = au(0), u'(1) = -au(1)\}. \quad (4)$$

Every $u \in \text{dom}(S)$ satisfies these boundary conditions, and the two operators agree there. Hence $S \subset A_a$ for every $a \geq 0$.

Since A_a is nonnegative self-adjoint, $\{e^{-rA_a}\}_{r \geq 0}$ is a strongly continuous contraction semigroup on H . It defines the propagator

$$U_a(t, s) = e^{-(t-s)A_a}, \quad 0 < s \leq t \leq T,$$

and the evolution semigroup on $L^p(\mathcal{I}, H)$

$$(e^{-\tau \mathcal{K}_a} f)(t) = e^{-\tau A_a} f(t - \tau) \mathbf{1}_{\mathcal{I}}(t - \tau). \quad (5)$$

Let $\mathcal{K}_a$ denote its generator in the sign convention of the source. Theorem 3.3 of [1] makes $\mathcal{K}_a$ an evolution generator. Alternatively, equation (5) is exactly the defining evolution-semigroup formula.

Let $\mathcal{A}_a$ be the multiplication operator induced by A_a . Lemma 3.8(ii) of [1] states that

$$\mathcal{K}_a f = D_0 f + \mathcal{A}_a f, \quad f \in \text{dom}(D_0) \cap \text{dom}(\mathcal{A}_a).$$

The inclusion $S \subset A_a$ implies $\text{dom}(S) \subset \text{dom}(\mathcal{A}_a)$ and $\mathcal{A}_a f = S f$ on $\text{dom}(S)$. Therefore

$$\tilde{\mathcal{K}} \subset \mathcal{K}_a \quad (a \geq 0).$$

Thus every $\mathcal{K}_a$ is a closed evolution-generator extension of the same evolution pre-generator $\tilde{\mathcal{K}}$.

It remains only to prove that these extensions are distinct. If $a \neq b$, take the cubic Hermite interpolant $u \in H^2(0, 1)$ determined by

$$u(0) = 1, \quad u'(0) = a, \quad u(1) = 0, \quad u'(1) = 0.$$

Then $u \in \text{dom}(A_a)$ but $u \notin \text{dom}(A_b)$, so $A_a \neq A_b$. If $\mathcal{K}_a = \mathcal{K}_b$, their evolution semigroups would coincide. By the one-to-one correspondence of Theorem 3.3 of [1], their propagators would coincide, so $e^{-rA_a} = e^{-rA_b}$ for every $0 < r \leq T$. Taking the strong derivative at $r = 0$ would give $A_a = A_b$, a contradiction. Hence the $\mathcal{K}_a$ are pairwise distinct. Since $[0, \infty)$ has cardinality continuum, the proof is complete. $\square$

4 Verification, scope, and novelty

The construction stays inside the exact framework of the source: H is separable; S is closed and densely defined; $\tilde{\mathcal{K}}$ has the canonical form (1); and every extension comes from a bounded strongly continuous propagator. The source itself emphasizes that its notion of solution operator is weak. The theorem answers that precise weak question; it does not claim classical well-posedness for the minimal-Laplacian equation. Indeed, Robin heat trajectories need not remain in $\text{dom}(S)$.

No computation is used in the proof. The two key checks are (1) that $S \subset A_a$ makes the full canonical domain of $D_0 + \mathcal{S}$, not just an algebraic tensor core, lie in every $\mathcal{K}_a$, and (2) that different Robin boundary conditions force different propagators and hence different evolution generators. The detailed audit is in `VERIFICATION.md`.

The local run indexes were searched for arXiv:1612.06147, “evolution pre-generator,” multiple or several evolution-generator extensions, nonunique solution operators, the minimal Laplacian, and Robin extensions. Bounded web searches on August 9, 2026 used the exact open-problem sentence, the arXiv id and DOI, and combinations of those terms. Exact-phrases searches returned the source paper; searches combining the terminology with Laplacian, Robin, and nonuniqueness found no later resolution. A citation-metadata check also inspected the titles of the small related Trotter/evolution-semigroup citation set and found no stated resolution. This is a bounded search, not a systematic bibliographic proof of novelty, so expert literature review remains necessary. Full search bounds are recorded in `novelty_search.md`.

References

- [1] H. Neidhardt, A. Stephan, and V. A. Zagrebnov, *Convergence rate estimates for Trotter product approximations of solution operators for non-autonomous Cauchy problems*, Publ. Res. Inst. Math. Sci. **56** (2020), no. 1, 83–135; arXiv:1612.06147; doi:10.4171/PRIMS/56-1-5.
- [2] T. Kato, *Perturbation Theory for Linear Operators*, second edition, Springer, Berlin, 1976.